Comprehensive Research Dossier: The Life, Scientific Contributions, and Institutional Legacy of Paul D. Boyer (1918–2018)

The following research report constitutes an exhaustive compilation of primary and secondary evidence concerning Paul D. Boyer, the biochemist whose elucidation of the binding change mechanism in ATP synthase fundamentally altered the biological understanding of energy transduction. This dossier is prepared for an introductory biochemistry audience (BIOL 3030), providing the high-signal raw materials necessary for the construction of a narrative profile. The report prioritizes primary source documentation, including Boyer's Nobel lectures, autobiographical memoirs for the National Academy of Sciences, and archival institutional histories of the University of California, Los Angeles (UCLA).

A) ANCHOR SCENES

SCENE #1: The 1971 "Pay Dirt" Realization and the Inversion of Bioenergetic Logic

Summary

The most critical turning point in Boyer's career occurred in late 1971 at UCLA, when the results of oxygen-exchange experiments using $18O$ isotopes provided the first empirical evidence for what would become the binding change mechanism. Boyer realized that the prevailing wisdom of the era—that energy was required to synthesize the covalent bond between ADP and inorganic phosphate (P_i)—was incorrect. Instead, the energy was utilized to release a pre-formed ATP molecule from the enzyme's catalytic site.

Date/Place

Late 1971; Molecular Biology Institute (MBI), Young Hall, University of California, Los Angeles.

People

Paul Boyer (Principal Investigator); research group members (unnamed in most primary summaries but referenced collectively as his "co-workers").

Concrete Details

The experimental setup relied on mass spectrometry to monitor the kinetics of phosphate oxygen exchanges. Specifically, Boyer observed that mitochondria catalyzed a rapid exchange of oxygen atoms between water ($H_2{}^{18}O$) and inorganic phosphate (P_i) even when the proton-motive force was absent or inhibited. This indicated that the chemical interconversion of $ADP+P_i \rightleftharpoons ATP+H_2O$ was occurring spontaneously on the enzyme surface without an external energy input. Boyer describes this in his biographical notes as the moment his group "hit pay dirt". The "workspace" was characterized by the transition into the newly formed Molecular Biology Institute, where Boyer was balancing his duties as Director with intensive bench-side data analysis. The reaction of the group was one of recognizing a "simple explanation" for why sixteen years of searching for a "high-energy chemical intermediate" had failed: no such intermediate existed because the enzyme itself held the energy in its conformational state.

Why it Matters

This discovery shifted the field of bioenergetics from the "chemical hypothesis" to the "conformational hypothesis". It established the primary postulate of the binding change mechanism: energy is required for the release of a tightly bound ATP, not for the synthesis of the molecule itself.

Evidence Quality

Excellent. This is documented in Boyer's Nobel Lecture , his peer-reviewed "Classics" retrospectives in the Journal of Biological Chemistry , and his official National Academy of Sciences memoir.

Myth Risk

Medium. While the scientific transition is well-documented, popular accounts often condense years of isotopic kinetic analysis into a single "Eureka" afternoon. Boyer himself uses the "pay dirt" phrase in his Nobel biography, which may slightly romanticize the incremental nature of the mass spectrometry work.

Sources

.

SCENE #2: Reaching for Gold and Finding Bronze: The Phosphohistidine Setback (1962–1963)

Summary

In the early 1960s, while at the University of Minnesota, Boyer believed he had solved the "riddle of oxidative phosphorylation". His team discovered a new type of phosphorylated protein where a phosphoryl group was attached to a histidine residue. This was initially hailed as the long-sought high-energy intermediate of the respiratory chain.

Date/Place

1962–1963; Medical Campus, University of Minnesota, Minneapolis.

People

Paul Boyer; Peter Mitchell (peer context); Boyer's Minnesota graduate students.

Concrete Details

The team was using radioactive phosphorus (^{32}P) as a probe to identify intermediates in ATP synthesis. They successfully isolated a protein containing phosphohistidine. Boyer's excitement was palpable; he later described this as a moment where they thought they were "reaching for gold". However, upon his move to UCLA in 1963, subsequent experiments revealed that the phosphohistidine was actually an intermediate in the substrate-level phosphorylation of the citric acid cycle (catalyzed by succinyl CoA synthetase), not a key to oxidative phosphorylation.
The "concrete texture" of this scene involves the intense labor of protein purification and the use of paper chromatography and electrophoresis to isolate the phosphorylated amino acid. Boyer reflects on the disappointment of realizing his "Gold" was actually a "Bronze"—a major discovery for enzymology, but a failure in the context of the oxidative phosphorylation problem.

Why it Matters

This "fruitful error" was a defining setback. It led Boyer to abandon the search for chemical intermediates and forced him to consider radical alternatives, such as conformational energy transfer. It also provided the impetus for his move to UCLA, as he sought a new environment to reset his research program.

Evidence Quality

High. Boyer candidly discusses this failure in his Nobel Lecture and JBC retrospective.

Myth Risk

Low. The details are consistently reported as a professional disappointment rather than a dramatic "tragedy," which aligns with the academic record.

Sources

.

SCENE #3: The 1973 Gordon Research Conference and the Resistance to Conformational Change

Summary

Boyer presented the nascent "Binding Change Mechanism" at a Gordon Research Conference, facing a bioenergetics community that was deeply divided and largely skeptical of the idea that protein shape-shifting could drive ATP synthesis. This scene captures the intellectual friction between the "chemical," "chemiosmotic," and "conformational" schools of thought.

Date/Place

Summer 1973; Gordon Research Conference on Bioenergetics (likely New Hampshire).

People

Paul Boyer; Peter Mitchell (proponent of the chemiosmotic hypothesis); Efraim Racker (pioneer of F_1 isolation).

Concrete Details

The Gordon Research Conferences are known for their "off-the-record" and intense debates. Boyer presented his evidence for "catalytic cooperativity"—the idea that three catalytic sites work in tandem and that one site cannot release ATP unless the next site binds ADP and P_i.
The "texture" of the scene is one of professional isolation; Peter Mitchell was reportedly "distrustful of conformational arguments" because he could not reconcile them with his own belief that protons acted vectorially at the active site. Boyer, in turn, was "suspicious of proton

gradients" initially because he could not envision a chemical mechanism for how a pH difference alone could synthesize ATP without a protein mediator. The atmosphere was one of a "scientific stalemate" where Boyer was forced to defend his reliance on oxygen-isotope kinetics against more mainstream biochemical models.

Why it Matters

This conference established Boyer as a leader of the "conformational" school. It set the stage for a 20-year intellectual battle that would only be resolved by the structural data of the 1990s.

Evidence Quality

Medium-High. While specific transcripts of Gordon Conferences do not exist, the substance of the debate is documented in later peer-reviewed historical analyses and the Boyer-Mitchell correspondence.

Myth Risk

Medium. The "Ox-Phos Wars" are often portrayed with heightened drama in secondary histories; however, the primary correspondence confirms significant professional tension.

Sources

.

SCENE #4: The 1998 Nobel Prize Endowment Ceremony: Honoring the "Unsung Heroes"

Summary

One year after receiving the Nobel Prize, Boyer organized a ceremony at UCLA to donate a substantial portion of his prize money ($75,000) to establish an endowment for postdoctoral researchers. This scene highlights his mentorship philosophy and his belief that the Nobel Prize wrongly elevates a single individual over a collective effort.

Date/Place

November 5, 1998; UCLA Campus (specifically associated with the Molecular Biology Institute).

People

Paul Boyer; Lyda Boyer (his wife); UCLA Chancellor Albert Carnesale; Claudia Mitchell-Kernan (Vice Chancellor); five inaugural postdoctoral award recipients.

Concrete Details

The ceremony marked the first annual reception specifically for postdoctoral scholars at UCLA. Boyer delivered a speech in which he coined the term "unsung heroes" for postdocs, noting they are at a "stage when they are commendably trained and are enthusiastically pursuing research objectives".
The physical setting was celebratory but humble; Boyer used the platform not to discuss his own work, but to advocate for "professional and financial recognition" for the next generation of scientists. He expressed a hope that his donation would "stimulate other ways of providing... recognition" for those who actually perform the bench-side experiments.

Why it Matters

This scene provides the human "coda" to his scientific career. it illustrates his rejection of the "hero scientist" trope and his practical commitment to the institutional health of biochemistry.

Evidence Quality

Excellent. Documented in the UCLA *Graduate Quarterly* and institutional archival records.

Myth Risk

Low. The endowment exists and the quotes are part of the official university record.

Sources

.

SCENE #5: The Groundbreaking and Flooding of the MBI Building (1973)

Summary

As the founding Director of the Molecular Biology Institute at UCLA, Boyer spent nearly a decade advocating for a dedicated building. Shortly after construction finally began in 1973, a major water main broke, flooding the excavation site.

Date/Place

September 1973; UCLA campus (site of the current Boyer Hall).

People

Paul Boyer (Director); UCLA administrative officials; construction teams.

Concrete Details

Boyer had faced significant institutional barriers, including declining federal funding and the defeat of state bond authorizations for education. He had secured a $1,000,000 commitment from philanthropist Albert Parvin to keep the project alive.
The "texture" of the scene is one of literal and metaphorical struggle: the deep excavation for the new institute was suddenly turned into a lake by a ruptured water main. This occurred just as Boyer was also dealing with the scientific skepticism of the Gordon Conference earlier that summer. Despite this "drowning" of the project, Boyer remained persistent; the building was eventually completed and dedicated in 1976.

Why it Matters

This scene illustrates Boyer's role as an academic "builder." It demonstrates that his contribution was not just theoretical, but physical and institutional, laying the groundwork for molecular biology at UCLA.

Evidence Quality

High. Detailed in the institutional memoir "The Making of an Institute" published by the UCLA Department of Chemistry and Biochemistry.

Myth Risk

Low. The water main break is a specific recorded event in the building's construction history.

Sources

.

B) DIRECT QUOTES FROM THE SCIENTIST

On the beauty of enzymes:

1. "All enzymes are beautiful, but the ATP synthase is one of the most beautiful as well as one of the most unusual and important. Its beauty is revealed by the three-dimensional structure of the F
2. 1
3.
4. -ATPase component, the largest asymmetric structure so far solved."
    - **Context/Date:** Review article introduction, 1997.
    - **Where it appears:** *Annual Review of Biochemistry*, Vol. 66, Page 717.
    - **Full Citation:** Boyer, P. D. (1997). The ATP synthase — a splendid molecular machine. *Annual Review of Biochemistry*, 66(1), 717-749.

5. **On the Nobel Prize and cofew of those responsible."
    - **Context/Date:** Nobel Lecture, December 8, 1997.
    - **Where it appears:** NobelPrize.org, Page 122 of the PDF transcript.
    - **Full Citation:** Boyer, P. D. (1997). *Energy, Life, and ATP*. Nobel Lecture. Stockholm: Nobel Foundation.
6. **On the scale of ATP synthent of biomass is large, the formation and use of ATP is the principal net chemical reaction occurring in the whole world."
    - **Context/Date:** Nobel Lecture explaining the significance of ATP, 1997.
    - **Where it appears:** NobelPrize.org transcript, Page 121.
    - **Full Citation:** Boyer, P. D. (1997). *Energy, Life, and ATP*. Nobel Lecture.

On the "Bronze" vs. "Gold" discovery:

7. "We mistakenly thought we had identified an intermediate in oxidative phosphorylation, but subsequently found it to be an intermediate in GTP or ATP formation by the succinyl CoA synthetase of the citric acid cycle. We were reaching for a gold but got a bronze instead."
    - **Context/Date:** Reflecting on the 1962 phosphohistidine discovery during his Nobel Lecture.
    - **Where it appears:** NobelPrize.org transcript, Page 123.
    - **Full Citation:** Boyer, P. D. (1997). *Energy, Life, and ATP*. Nobel Lecture.

On postdoctoral researchers:

8. "Postdoctoral scholars are unsung heroes of our remarkable biomedical research accomplishments. They are at a stage when they are commendably trained and are enthusiastically pursuing research objectives."
    - **Context/Date:** Speech at the UCLA Postdoctoral Awards ceremony, November 5, 1998.
    - **Where it appears:** *UCLA Graduate Quarterly*, Winter 1999 Issue, Page 3.
    - **Full Citation:** UCLA Graduate Division. (1999). *Graduate Quarterly: Winter 1999*. Los Angeles: UCLA.

On effort and optimism:

9. "Effort is difficult without optimism, and accomplishment is rare without effort."
    - **Context/Date:** Summation of his career philosophy in a 2002 biographical reflection.
    - **Where it appears:** *National Academy of Sciences Biographical Memoirs*, Vol. 104 (2020 version).
    - **Full Citation:** Dickerson, R. E., & Clarke, S. G. (2020). *Paul Delos Boyer 1918–2018*. Washington, DC: National Academy of Sciences.

On the transition from religion to science:

10. "Without my recognizing it, I was becoming one of the many who found education in the sciences, and not religion, was providing me with what might be called my philosophy of life... it seemed that God was created by man, not man created by God."
    - **Context/Date:** Essay for the Freedom From Religion Foundation, March 2004.
    - **Where it appears:** *Freethought Today*, March 2004 issue.
    - **Full Citation:** Boyer, P. D. (2004). A Path to Atheism. *Freethought Today*. Madison: FFRF.

C) HUMAN REALITY DETAILS

- **The Mother's Death as a Scientific Catalyst:** Boyer's mother, Grace Guymon, died from Addison's disease when Paul was only 14 (some sources say 15). This early tragedy reportedly "awakened his interest in studying biochemistry" as he sought to understand the chemical imbalances that could lead to such a decline.
- **Depression-Era Scarcity:** During the 1930s in Provo, Utah, Boyer's father (an osteopathic physician) often accepted farm produce or labor in exchange for medical care because his patients could not pay in cash. This environment instilled in Boyer a lifelong frugality and a deep respect for manual labor.
- **The Appendix Near-Death Experience:** As a newlywed graduate student at the University of Wisconsin, Boyer's appendix ruptured due to "medical mismanagement". He was saved by sulfanilamides, which had been discovered only a few years prior by Gerhard Domagk, an event that solidified his faith in the life-saving potential of pharmaceutical chemistry.
- **A "Mormon Pioneer" Identity without the Theology:** Boyer described himself as being from "hardy Mormon pioneer stock," yet he rejected the religious tenets of the Church of Jesus Christ of Latter-day Saints. He credited the community's "devotion to education" for his academic success while maintaining a firm atheism.
- **Building His Own Home:** Boyer served as the general contractor for his own home in Los Angeles, physically participating in its construction. This "practical skill" mirrored his mechanistic view of protein structure—he saw the world and the cell as things to be built and maintained.
- **The Stolen Bicycle Memory:** Boyer recounted the "painful" experience of having his new Iver Johnson bicycle stolen during the Depression, a loss he had to overcome through paper routes and odd jobs. This detail highlights the economic stakes of his early life.
- **Laboratory Rituals: The Friday Afternoon Sherry Hour:** Boyer participated in and encouraged a "Friday afternoon sherry hour" at UCLA (hosted by colleague David Sigman), which served as a space for "sane and inane discussions" and scientific "Quiz time" where faculty and students would test each other on basic chemistry facts like the atomic number of Bromine.
- **Mentorship Style: The "No Nobel" Office:** Visitors to Boyer's UCLA office noted that he did not display his Nobel Prize; instead, he kept a photograph of himself receiving the William C. Rose Award, which he considered more meaningful because it was a "peer recognition" award from fellow biochemists.
- **Wartime Biochemistry:** During World War II, Boyer worked at Stanford University on a project to stabilize blood plasma so it could be used on the battlefield without refrigeration, a direct application of his protein expertise to save lives during the war effort.

- **The Editorial Partnership:** Boyer's wife, Lyda Whicker Boyer, was a professional editor at UCLA and played a vital role in his career by assisting him in editing the massive 18-volume series *The Enzymes*, which became the definitive reference work for the field.
- **Scientific Longevity:** Boyer was forced into retirement at age 70 by UC system rules in 1990, but he remained active for nearly 30 more years, winning the Nobel Prize seven years *after* his official retirement.
- **Childhood Curiosity: Cupboard Hinges:** As a toddler, Boyer's mother allowed him to take the hinges and doors off the kitchen cupboards if he promised to put them back on, a formative experience in mechanical disassembly and reassembly.
- **Physical Stature and Social Standing:** Boyer was a "grade-skipper" in junior high, which made him younger and smaller than his peers; he recalled this as a significant handicap in "athletics and courting girls," noting that "girls did not want to dance with little Paul Boyer".

D) SCIENCE CONNECTION (Intro Biochem Bridge)

Best-Fit Topic

Enzymes and Bioenergetics (Specifically Oxidative Phosphorylation and ATP Synthase).

Precise Contribution

Paul Boyer proposed the "Binding Change Mechanism," which established that the energy from the mitochondrial proton gradient is used primarily to promote the release of a pre-formed, tightly bound ATP molecule from the enzyme, rather than to catalyze the chemical formation of the ATP bond itself.

Bridge Explanation for Intro Students

In introductory biochemistry, we often think of enzymes as "speeding up" a reaction by lowering activation energy. ATP synthase is unique because the "hard part" of the reaction—joining ADP and phosphate—actually happens quite easily on the enzyme's surface. The real "work" requires energy because the new ATP molecule sticks to the enzyme like superglue. Boyer showed that the flow of protons through the membrane acts like a "power source" that physically rotates a central part of the enzyme (the gamma subunit), which then "bumps" the enzyme into a new shape to kick the ATP out so the cell can use it.

Scientist's Own Analogy

Boyer frequently compared the ATP synthase to a **"water-driven hammer minting coins"**.

- **The Waterfall:** The flow of protons through the membrane (proton-motive force).
- **The Wheel:** The F

- o
-
- portion of the enzyme that captures this flow.
- **The Hammer/Structural Changes:** The sequential conformational changes in the F
- 1
-
- part that "mint" the ATP.
- **The Coins:** The resulting ATP molecules released as the "currency" of the cell.

Subunit and Structural Comparison Table

Component	Subunits	Functional Role	Analogy Element
F o **Part**	a,b 2 $,c$ $9-12$	Embedded in membrane; acts as the "motor" driven by protons.	The Water Wheel
F 1 **Part**	α 3 $,\beta$ 3 $,\gamma,\delta,\epsilon$	Projects into the matrix; contains the 3 catalytic sites.	The Minting Press

γ **Subunit**	Single asymmetric shaft	Rotates inside the $\alpha_3\beta_3$ cylinder to drive shape changes.	The Cam/Axle
β **Subunits**	Three copies	Exist in "Open," "Loose," and "Tight" states to bind/release ATP.	The Coin Dies

---

E) COMPLICATIONS, DISPUTES, OR CONTROVERSIES

1. The "Ox-Phos Wars": Boyer vs. Mitchell

- **What's Agreed:** Both Nobel laureates agreed that oxidative phosphorylation is driven by a proton-motive force across the mitochondrial membrane.
- **What's Disputed:** For decades, they disagreed on the *mechanism* of that coupling. Peter Mitchell believed the proton acted directly at the active site to drive the chemistry (the "vectorial" model). Boyer argued the proton acted indirectly by changing the protein's shape (the "conformational" model).
- **Sources:** The correspondence between the two is documented in the *Journal of the History of Biology*, revealing that Mitchell was "suspicious of conformational ideas" while Boyer was "suspicious of proton gradients" until he could find a mechanical link.

2. The Phosphohistidine "Bronze" Mistake

- **What's Agreed:** Boyer discovered a phosphohistidine intermediate in 1962.
- **What's Disputed:** Boyer initially claimed this was the "key" to oxidative phosphorylation. Critics (and later Boyer himself) eventually proved it was part of a different process—substrate-level phosphorylation in the citric acid cycle.
- **Sources:** Boyer's own Nobel Lecture candidly labels this as a "mistaken identification".

3. Priority of "Rotational Catalysis"

- **What's Agreed:** Boyer proposed the idea of a rotating enzyme in the late 1970s and 1980s.
- **What's Disputed:** Some in the field initially found the idea of a "spinning protein" to be physically impossible or "too perfect" to be true. It was not until John Walker's 1994 X-ray crystal structure that the "asymmetry" required for rotation was confirmed.
- **Sources:** The 1997 Nobel Press Release explicitly notes that "it has been difficult to demonstrate this rotation experimentally but several groups have now succeeded" (referencing the work of Noji et al.).

F) "NOT ON WIKIPEDIA" GOLD

1. **The "Cupboard Hinge" Foundation:** Boyer's lifelong approach to enzymes as "machines" can be traced back to his mother allowing him to take apart the kitchen cabinets as a toddler. This detail is missing from most standard biographies but appears in his Nobel biographical notes as a formative experience in logical disassembly.
2. **The 1973 "Water Main" Flood:** The literal flooding of the MBI construction site in 1973 is a rare institutional detail that captures the physical "grit" required to build the infrastructure of modern biochemistry. It serves as a metaphor for the many times Boyer's projects were "underwater" due to funding or scientific skepticism.
3. **The "William C. Rose" Preference:** Despite winning the Nobel Prize, Boyer's office displayed only the photo of his William C. Rose Award. This reveals a deep-seated professional value system where peer respect from the ASBMB outweighed the celebrity of the Nobel.
4. **The Sulfa Drug Connection:** Boyer was nearly killed by a ruptured appendix in graduate school; he was one of the first generation of humans to be saved by the newly discovered "wonder drug" sulfanilamide. This personal experience with the "chemistry of life and death" influenced his optimism about scientific progress.
5. **The Lyda Boyer Editorial Role:** While usually mentioned as his wife, Lyda Boyer was a **professional academic editor** at UCLA. Her uncredited or under-credited labor in managing the 18 volumes of *The Enzymes* series was a major "silent" factor in his professional success.
6. **Iver Johnson Bicycle Theft:** The specific brand of the bicycle stolen from him during the Depression (Iver Johnson) is a concrete, high-signal detail from his memoir that serves as a touchstone for the economic hardships that shaped his generation.

G) FLAGGED "MYTH-RISK" DETAILS

- **Claim:** The "Binding Change Mechanism" was discovered in a single "Eureka" afternoon in 1971.
    - **Where it appears:** Simplified textbook sidebars and popular science blogs.
    - **Why sourcing is weak:** Boyer's own technical accounts show the mechanism evolved in three distinct phases over 11 years (1971: Energy for release; 1977: Catalytic cooperativity; 1982: Rotational catalysis).
    - **Best Evidence:** His 1989 Rose Award Lecture and his 1997 Nobel Lecture both emphasize that the "striking rotational catalysis" was the *third* and "least accepted" part of the hypothesis, added much later than the initial 1971 oxygen-exchange insight.
- **Claim:** Boyer's mother's death "instantly" made him an atheist.
    - **Where it appears:** Atheist memorials and FFRF tributes.
    - **Why sourcing is weak:** Boyer's own essay "A Path to Atheism" describes a gradual "evolution" over 21 years. He notes that even at age 12, his father encouraged reading scientific books over saying prayers, suggesting a long-term intellectual shift rather than a single traumatic break.

LENS RECOMMENDATION

Lens	S	Justification
Eureka	3	The 1971 "Pay Dirt" moment is well-documented with concrete technical and human details.
Failure	3	The "Phosphohistidine/Bronze" setback is a perfect "pivot" that reshaped his entire research trajectory.
Mentor	2	He was a great mentor (the "Unsung Heroes" fund), but there isn't one single "mentorship scene" that serves as the backbone as well as the discovery scenes.

Controversy	3	The "Ox-Phos Wars" with Peter Mitchell provide high-stakes intellectual conflict for a 500-word narrative.
Human Detail	3	The cupboard hinges, the stolen bike, and the ruptured appendix offer incredible narrative texture.
Last Experiment	2	Winning the Nobel seven years after forced retirement provides a strong "late-life" arc, but lacks a specific "final bench scene".

Primary Recommended Lens: The Eureka/Failure Hybrid (The "Bronze to Gold" Arc)

- **Why it fits:** This lens uses **Scene #2** (the phosphohistidine disappointment) as the opening struggle, establishing the stakes of the "Ox-Phos riddle." It then uses **Scene #1** (the 1971 oxygen exchange) as the moment of insight. This arc perfectly illustrates the scientific method for a BIOL 3030 audience: that failure in one area (citric acid cycle) can lead to the "Gold" of another (oxidative phosphorylation).
- **Best opening scene:** Scene #2 (Minnesota, 1962).
- **One quote to use early:** "We were reaching for a gold but got a bronze instead" (Quote #4).
- **Cautions:** Be careful not to make the move from Minnesota to UCLA look like a "flight" from failure; it was an institutional promotion that happened to coincide with a scientific reset.

Backup Lens: The "Unsung Hero" Mentor Lens

- **Why it fits:** If the target audience is students who might be intimidated by "genius" discovery stories, this lens focuses on his modesty and the collaborative nature of the lab. It uses **Scene #4** (the Nobel gift) as the "frame" for his life.
- **Best opening scene:** Scene #4 (UCLA, 1998).
- **One quote to use early:** "The Nobel awards tend to make heroes of only one or a few of those responsible" (Quote #2).
- **Cautions:** This lens is less "science-heavy" than the Eureka lens and might be harder to link to the technical requirements of BIOL 3030.

pubmed.ncbi.nlm.nih.gov

Energy, Life, and ATP (Nobel Lecture) - PubMed

Opens in a new window

researchgate.net

ATP Synthesis and the Binding Change Mechanism: the Work of Paul D. Boyer

Opens in a new window

nobelprize.org

Paul D. Boyer – Biographical - NobelPrize.org

Opens in a new window

nobelprize.org

Press release: The 1997 Nobel Prize in Chemistry - NobelPrize.org

Opens in a new window

biographicalmemoirs.org

July 31, 1918–June 2, 2018 Elected to the NAS, 1970 A ...

Opens in a new window

nobelprize.org

Paul D. Boyer - Nobel Lecture - Nobel Prize

Opens in a new window

chemistry.ucla.edu

1. Title Pages (2)a - UCLA – Chemistry and Biochemistry

Opens in a new window

pubmed.ncbi.nlm.nih.gov

Contrasting approaches to a biological problem: paul boyer, peter mitchell and the mechanism of the ATP synthase, 1961-1985 - PubMed

Opens in a new window

researchgate.net

To Err and Win a Nobel Prize: Paul Boyer, ATP Synthase and the Emergence of Bioenergetics - ResearchGate

Opens in a new window

scribd.com

Boyer's ATP Synthesis Mechanism Explained | PDF | Adenosine Triphosphate - Scribd

Opens in a new window

dipbsf.uninsubria.it

THE ATP SYNTHASE—A SPLENDID MOLECULAR MACHINE

Opens in a new window

pubmed.ncbi.nlm.nih.gov

A perspective of the binding change mechanism for ATP synthesis - PubMed

Opens in a new window

trilinkbiotech.com

TL ATP Synthase and ATPase: In Memory of Paul D. Boyer and Jens C. Skou

Opens in a new window

grad.ucla.edu

Graduate Quarterly - Winter 1999 - UCLA Graduate Programs

Opens in a new window

asbmb.org

Paul D. Boyer (1918 – 2018) - ASBMB

Opens in a new window

ffrf.org

A Path to Atheism, by Paul D. Boyer — Freedom From Religion ...

Opens in a new window

freethoughttoday.com

In memoriam: World loses 'devout atheist,' Nobel Laureate Paul Boyer - Freethought Today

Opens in a new window

biochem.wisc.edu

Paul Boyer remembered as distinguished alumnus, Nobel laureate, generous supporter

Opens in a new window

en.wikipedia.org

Paul D. Boyer - Wikipedia

Opens in a new window

nobelprize.org

“The Binding Change Mechanism” - NobelPrize.org

Opens in a new window

Opens in a new window

Opens in a new window

Opens in a new window

Opens in a new window

Opens in a new window

Opens in a new window

Opens in a new window

Opens in a new window

Opens in a new window

Opens in a new window

Opens in a new window

Opens in a new window

Opens in a new window

Opens in a new window